



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER II 2024/2025

(SECP2753) DATA MINING

SECTION 01

ASSIGNMENT 1

LECTURER: PROF. MADYA DR. ROLIANA BINTI IBRAHIM

STUDENT NAME	MATRIC NO
GUI KAH SIN	A23CS0080
POH LOK YEE	A23CS0262

Insights and Preprocessing on Abalone and Liver Disease Datasets

1.0 Introduction

Data mining enables us to extract meaningful insights from large amounts of datasets and is also known as Knowledge Discovery in Databases (KDD). In any data mining or machine learning task, data preprocessing is an essential step that ensures the quality and reliability of the final results. Raw data often contains inconsistencies, missing values, duplicates, or noise and outliers that need to be addressed before applying analytical techniques or even building predictive models. In this assignment, we mainly focus on two datasets, which are abalone and liver disease from Kaggle and the UCI Machine Learning Repository. This assignment begins with an overview and insights about those two datasets, highlighting their structure and key characteristics. Following the overview, one dataset will be selected for proposed detailed data preprocessing, including steps like data transformation (which is normalization), and data discretization. The aim is to produce a high-quality dataset that can be used for effective analysis and decision-making.

2.0 Dataset Overview

2.1 Abalone Dataset

The Abalone dataset consists of 4,188 instances, each characterized by eight features, also referred to as attributes, and one target variable aimed at predicting the age of the abalone. These attributes include a combination of categorical, continuous, and integer variables. The target variable is the number of rings on the abalone's shell, which can be used to estimate the age of abalone by adding 1.5 to the ring count. The measurements are recorded in either millimeters or grams, depending on the type of attribute.

Attribute	Data type	Description	Units	Min	Max
Sex	Categorical (Nominal)	M, F, and I (infant)	-	F (1307)	M (1528)

Length	Continuous	Longest shell measurement	mm	0.075	0.815
Diameter	Continuous	Perpendicular to the length	mm	0.055	0.650
Height	Continuous	With meat in the shell	mm	0	1.130
Whole_weight	Continuous	Weight of whole abalone	grams	0.002	2.825
Shucked_weight	Continuous	Weight of meat only	grams	0.001	1.488
Viscera_weight	Continuous	Gut weight after bleeding	grams	0.000	0.760
Shell_weight	Continuous	Weight after being dried	grams	0.002	1.005
Rings	Integer	Number of shell rings $\text{Age} \approx \text{Rings} + 1.5$	grams	1	29

From our data observation, we discovered several essential patterns in the Abalone dataset:

1. Sex (male, female and infant) and Size (length, diameter, and height):

Findings: Different sex of abalones may cause variations in size.

Insights: Female abalones tend to have larger size on average compared to male and infant abalones.

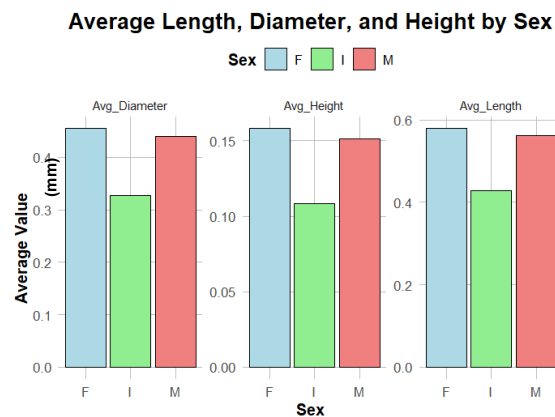


Figure 1: Relationship between Sex and Average Size (Length, Diameter, and Height)

2. Age and number of abalones

Findings: The age of abalones is determined by adding 1.5 years to the number of growth

rings. Based on this method, the age range of abalones spans from 2.5 to 30.5 years.

Insights: Most abalones are between 5 to 15 years old.

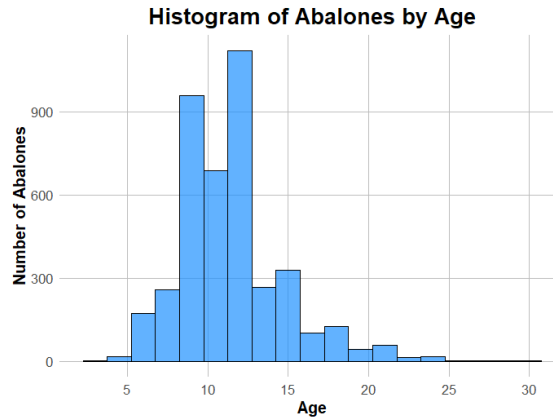


Figure 2: Relationship between Age and Number of Abalones

2.2 Liver Disease Dataset

The liver disease dataset contains 345 biochemical test results of blood samples from individuals potentially related to liver disorders caused by alcohol consumption. It consists of 5 clinical attributes which are mcv, alkphos, sgpt, sgot, gammagt while the 6th attribute, drinks, are the daily alcohol intake and can be used as a target variable for classification tasks. Below is a brief description and exploration of the insights derived purely from the variables provided:

Attribute	Data type	Description	Units	Min	Max
mcv (Mean corpuscular volume)	continuous	Average volume of a red blood cell	fl	65	103
alkphos (Alkaline phosphatase)	continuous	Liver enzyme that high levels may indicate liver disease	IU/L	23	138
sgpt (Alanine aminotransfases)	continuous	Liver enzyme that elevated in liver damage	IU/L	4	155
sgot	continuous	Liver enzyme that linked to	IU/L	5	82

		heart and muscle			
Gammagt (Gamma-GT)	continuous	Enzyme indicative of alcohol use and liver damage	IU/L	5	297
drinks	integer	Represent the number of alcoholic drinks consumed per day	unit	0	20

From our data observation, we discovered several essential patterns in the Liver dataset:

1. Gamma-GT level and drinks

Findings: There is a moderate positive correlation between daily alcohol consumption (drinks) and Gamma-GT levels which means individuals consume more alcohol drinks per day, their Gamma-GT levels tend to increase.

Insights: Higher alcohol consumption tends to be associated with higher Gamma-GT enzyme levels, which aligns with medical expectations since Gamma-GT enzyme is sensitive to liver stress caused by alcohol.

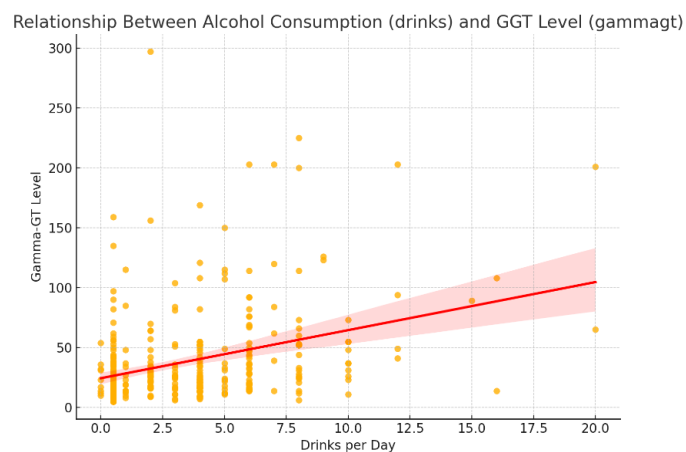


Figure 3: Relationship between Alcohol Consumption (drinks) and Gamma-GT Level (gammagt)

2. Sgot and drinks

Findings: As the alcohol drinks consumption increases, the sgot levels tend to rise. This prove that alcohol has noticeable impact on liver function, causing stress or damage reflected in higher sgot

values.

Insights: Sgot provides additional evidence of alcohol's impact on liver and potentially other tissues. Elevated sgot levels typically indicate liver stress or damage, which can result from chronic alcohol drinks consumption.

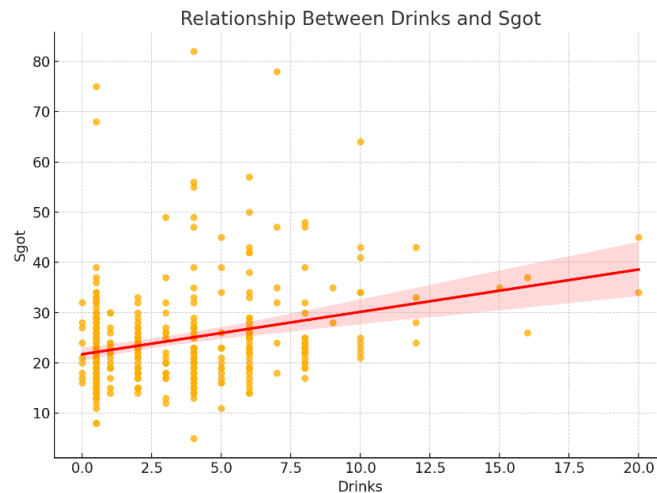


Figure 4: Relationship between Drinks and Sgot

3.0 Data Preprocessing for Abalone

The Abalone dataset was selected for preprocessing tasks, particularly focusing on normalization and discretization techniques. These techniques are important to ensure that the dataset is in the best possible shape for machine learning tasks. Compared to the Liver dataset, the Abalone dataset provides a richer set of continuous numerical attributes such as Length, Diameter, Height and various weight measurements. These attributes vary in range and scale, making the dataset ideal for demonstrating normalization processes. Furthermore, the Rings attribute, which is used to estimate the age of abalones, presents a valuable opportunity for discretization.

3.1 Normalization

In preparation for data mining model development, normalization was applied to the continuous numerical variables in the abalone dataset. The purpose of normalization is to transform all feature values into a common range, typically within the range of 0 to 1. This process is crucial because the dataset contains variables with varying units and magnitudes.

In the Abalone dataset, the WholeWeight attribute represents the weight of the abalone including its shell. This variable has a wide range with values approximately from 0.1 to 2.0. Without normalization, this large scale can dominate other variables like Height, which only ranges from 0.05 to 0.2.

To ensure that each feature contributes equally during model training, min-max normalization was applied. The following variables were normalized: Length, Diameter, Height, WholeWeight, ShuckedWeight, VisceraWeight, and ShellWeight.

The formula used for normalization is:

$$\text{Normalization, } v' = \frac{v - \min(A)}{\max(A) - \min(A)} (new_{\max(A)} - new_{\min(A)}) + new_{\min(A)}$$

Where:

- v is the original value of the attribute
- $\min(A)$ and $\max(A)$ are the minimum and maximum values of the attribute A
- $new_{\max(A)}$ and $new_{\min(A)}$ are the new scale limits which typically 0 and 1

For example, we apply this formula to the WholeWeight attribute, which ranges from $\min(A) = 0.002$ to $\max(A) = 2.2855$, and we want to normalize the value $v = 0.514$ into the range $[new_{\min(A)} = 0, new_{\max(A)} = 1]$, the result is:

$$v' = \frac{0.514 - 0.002}{2.2855 - 0.002} (1 - 0) + 0 = 0.181$$

This normalized value ensures WholeWeight will be on the same scale as other features, reducing the risk of dominance by high-range variables during model training.

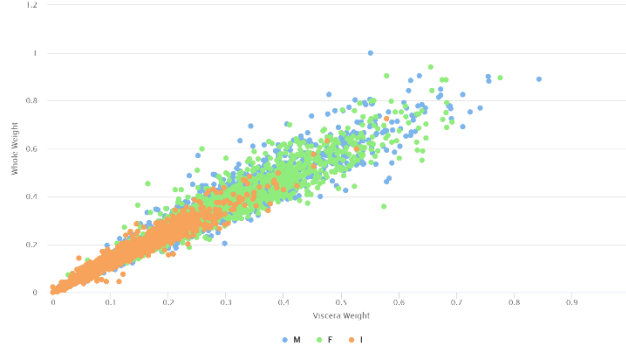


Figure 5: Relationship between Whole Weight and Viscera Weight by Sex of Abalone After Normalization

3.2 Discretization

Discretization is a crucial data preprocessing technique where continuous numerical data is transformed into categorical data by dividing the range of the attribute into a finite number of intervals or “bins.” This step is particularly important when working with classification models that perform better with categorical variables or when we want to group numerical attributes for interpretability.

For abalone dataset, we performed discretization on the Rings attribute in the Abalone dataset. The Rings variable is a continuous numeric feature ranging from 2.5 to 30.5, and it is typically used to estimate the age of an abalone. The formula used to approximate the abalone’s age is:

$$Age = Rings + 1.5$$

However, instead of using the continuous values directly, we opted to categorize them into 3 distinct age groups. This makes the analysis easier to understand, allows visual grouping, and can assist classification models by reducing complexity. To determine the bin width, the below steps was carried out:

$$\text{Total range} = 30.5 - 2.5 = 28$$

$$\text{Number of bins} = 3$$

$$\text{So, bin width} = 28 \div 3 = 9.3$$

Therefore, we could determine the bin ranges with labels:

Bin No	Range (Age)	Age Groups
1	2.5 – 11.8	Young
2	11.8 – 21.2	Middle-aged
3	21.2 – 30.5	Old

For example, an abalone with 15 rings will be 16.5 years old. This means it falls between 11.8 and 21.2. This abalone will be considered middle-aged.

The Age Group Distribution of Abalones

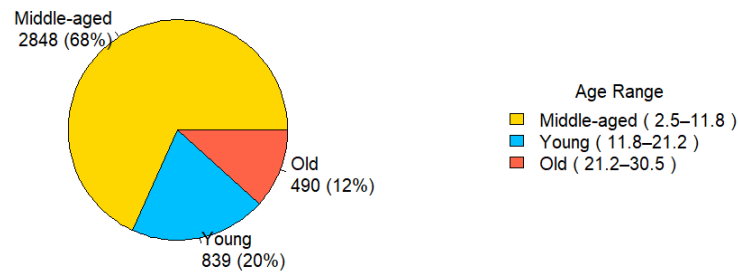


Figure 6: The Age Group Distribution of Abalones After Discretization Process

4.0 Discussion

Data preprocessing is a vital step in data mining that involves transforming raw data into a suitable format for analysis. This includes techniques like normalization and discretization to enhance data quality, improve model performance, and ensure fair comparisons across features.

Normalization is the process of scaling numerical features to a common range, typically from 0 to 1 ($[0,1]$). This is essential when attributes vary in magnitude and units, as it prevents features with large values from dominating those with smaller range. In the Abalone dataset, normalization was applied to attributes like Length, Diameter, Height, and those various weight features. This could ensure that each attribute contributed equally during model training, particularly for algorithm sensitive to distances such as k-NN or SVM.

While discretization involves converting continuous data into categorical data bins. This is useful for simplifying data, reducing noise, and enabling the use of classification models that perform better with categorical inputs. In this dataset, the Rings attribute, which represent age, was

discretized into three age groups: Young (2.5 - 11.8), Middle-aged (11.8 - 21.2), and Old (21.2 – 30.5). The age of an abalone is estimated by adding 1.5 to the number of rings, thus this transformation made the data more interpretable and aligned with the natural distribution of abalone ages.

5.0 Conclusion

Effective data preprocessing is a foundational step in any data mining or machine learning workflow. By examining the Abalone and Liver Disease datasets, we gained valuable insights into how raw data can be cleaned, transformed, and prepared for analysis. The Abalone dataset was selected for detailed preprocessing as it has rich numeric attributes, which provided an ideal opportunity to apply normalization and discretization. It showcased the importance of normalization for handling attributes with different scales, and discretization for simplifying continuous variables like age into meaningful categories.

These preprocessing steps ensured that all numerical features contributed equally during model training, which helps in reducing the potential bias from different value scales and enhanced data interpretability. Overall, the applied techniques improved data quality and prepared the dataset for accurate and reliable analysis in future modeling tasks.

6.0 References

1. Nash, W., Sellers, T., Talbot, S., Cawthorn, A., & Ford, W. (1994). *Abalone [Dataset]*. UCI Machine Learning Repository. <https://doi.org/10.24432/C55C7W>.
2. *Abalone Dataset*. (2018, July 19). Kaggle. <https://www.kaggle.com/datasets/rodolfomendes/abalone-dataset/data>
3. *liver disorders*. (2024, January 28). Kaggle. <https://www.kaggle.com/datasets/fatemehmehrpavar/liver-disorders>
4. *Liver Disorders [Dataset]*. (2016). UCI Machine Learning Repository. <https://doi.org/10.24432/C54G67>.